

FIELD MANUAL

Master Chess

Turn your own games into your training plan.

DIAGNOSE ♦ **EXPLAIN** ♦ **PRESCRIBE**

An adaptive educational chess tutor

Player & Mechanics Manual • Version 1.0

How to read this manual

This manual is written the way a good board-game rulebook is written: the rules first, then a fully worked *game of play* that walks one real game from upload to a finished training plan, so you can see every mechanism act on real moves. Wherever you see a tinted box, it is doing one of three jobs.

EXAMPLE OF PLAY — HOW THESE BOXES WORK

Boxes like this one follow a single running example — the analysis of a real historical game — so that every abstract rule is immediately grounded in something concrete. Read them in order and you will have watched the whole system operate end to end.

■ Rule box — the exact mechanic

Boxes like this state a precise rule: a threshold, a formula, a piece of scheduling logic. These are the authoritative numbers, taken directly from the system's source of record.

★ NOTE

Boxes like this flag something easy to get wrong, a limitation, or a cross-reference worth keeping in mind.

THE ONE IDEA TO KEEP

Master Chess is not an engine with a chat window bolted on. It runs a **closed learning loop** — diagnose what you can and cannot do, explain it against engine truth, prescribe exactly what to study and drill — and **every score it shows you has a receipt**: the exact moves that produced it. No black box.

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1 What Master Chess Is

Master Chess is an adaptive educational chess tutor. You upload your own games; it analyses every move with the Stockfish engine, and from that engine-grounded analysis it builds a persistent, *explainable* model of your ability across a taxonomy of **27 skills**. It then diagnoses your playing **level** and your current **plateau**, writes you a personalised **training plan** with matched reading, and turns the very mistakes you made into spaced-repetition **drills**.

The product's thesis is that expert chess knowledge should be encoded into a structured substrate, and instruction rendered dynamically for the individual learner from engine truth — never from an engine's chat-layer guesswork.

1.1 The two learning loops (and why the AI is neither)

The system has exactly two places where “learning” happens, and a language model is not one of them:

1. **The Player Model** — a structured representation of you across 27 skills, updated by Bayesian Knowledge Tracing from engine-verified evidence in every game and drill.
2. **The Chess Knowledge Layer** — engine ground truth, an opening / master-game library, and a book corpus mapped to the skill taxonomy.

An **AI reasoning layer** sits on top to write prose explanations, but it is always subordinate to engine truth: a runtime *claim guard* rejects any generated sentence that contradicts a verified move or evaluation. If the model cannot say something true, the app falls back to a deterministic, engine-only readout rather than let an unverified sentence reach you.

■ What grounds every claim

The engine (Stockfish) and the classified move record are the source of truth. The player model is derived from them by fixed, inspectable rules. The language model may only *phrase* facts it is handed; it may never introduce a move, number, or verdict of its own. Any sentence that fails verification is discarded.

1.2 Where the AI is used — and where it isn't

Master Chess uses AI deliberately and narrowly. Everything that *scores, judges, or decides* is deterministic — the engine and fixed rules. The AI is a **language layer** that turns those verified facts into readable coaching, and answers your questions. It appears in exactly four places:

Where	What the AI does	Grounded by
Game Review summary	Writes the post-game coaching note from your result, accuracy, and worst move	Claim guard + deterministic fallback
Prophylaxis notes	Phrases <i>why</i> a defensive move mattered	Deterministic probe decides the score; AI only narrates
Progress page	Writes your “coach’s summary” from your real stats	Real numbers only + deterministic fallback
Help assistant	Answers “how do I...” questions in chat	Fixed knowledge base; refuses out-of-scope

In every case the same rule holds: **the AI may only rephrase what the engine and your data already establish.** It never sets a mastery number, never classifies a move, never chooses your next action (a fixed rule does that), never invents chess advice, never generates a drill hint (those are derived from the engine’s best move), and **never plays your opponent in Play vs AI — that is Stockfish.** If the AI backend is unavailable, every one of these surfaces degrades gracefully to a deterministic, engine-only version — the app keeps working, just in plainer prose. The AI provider is pluggable (a self-hosted model by default), so no third-party service is required for the core loop.

1.3 How it runs

Layer	Technology
Client	React single-page app served by Vite (development port 5173)
Server	Express API on tsx (default port 8030)
Storage	SQLite via Drizzle ORM (better-sqlite3, WAL mode)
Engine	Stockfish over UCI, pooled; default search depth 16, multi-PV 4
Reasoning	Provider-abstracted LLM (local Ollama by default; hosted models by tier)

To run it locally: `npm run dev` starts client and server together; the server first applies database migrations, seeds the master-game library, resumes any interrupted analyses, and purges expired sessions. A health probe lives at `GET /api/health`.

★ SCOPE NOTE

This manual documents the **Master Chess** standalone app (the `ai_chess` project). A sibling project in the same workspace is a separate classroom product with teacher/student dashboards; those screens are *not* part of Master Chess and are not described here.

2 The Vocabulary: Skills, Levels, Plateaus

Everything the app says about you is expressed in three linked vocabularies. Learn these and the rest of the manual reads easily.

2.1 The 27 skills, in 4 categories

Each skill carries a mastery score from 0–100, a trend arrow, and — once there is evidence — a list of the exact moves that produced the score. Three skills marked *partial* are only partially diagnosable from game data alone.

#	Skill	Diagnosable
OPENING		
1	Opening principles	yes
2	Opening repertoire	yes
3	Opening preparation & novelties	partial
MIDDLEGAME		
4	Strategic planning	yes
5	Positional play	yes
6	Patience & long-term thinking	yes

#	Skill	Diagnosable
7	Prophylaxis	yes
8	Tactical pattern recognition	yes
9	Tactical consistency (blunder prevention)	yes
10	Calculation precision	yes
11	Attack & initiative	yes
12	Piece activity & coordination	yes
13	Defence & counterplay	yes
14	Converting advantages	yes
ENDGAME		
15	Endgame principles	yes
16	Game transition (middlegame to endgame)	yes
17	Endgame precision & conversion	yes
18	Pawn endings	yes
19	Rook endings	yes
20	Knight endings	yes
21	Bishop & mixed-piece endings	yes
PSYCHOLOGY & MENTAL		
22	Thought process & candidate moves	partial
23	Pattern recognition speed	yes
24	Intuition & practical judgment	partial
25	Time management	yes
26	Psychological resilience	yes
27	Psychological warfare	partial

★ ON THE COUNT

The taxonomy contains **27** skills across the four categories, and the app tracks and displays all 27. (How each one is scored is covered in §4.4.)

2.2 The 7 player levels

Your level is derived only from the rating carried in your games' PGN headers (Chess.com and Lichess exports include it). Upload unrated games and the app honestly shows no level rather than guessing one.

ID	Name	Rating	Dominant failure mode
L1	Newcomer	0--799	Tactical invisibility --- misses what was just threatened
L2	Beginner	800--1199	Pattern blindness --- misses tactics beyond one move
L3	Casual club player	1200--1499	Contextual blindness --- misses in game what they solve in puzzles
L4	Improving club player	1500--1799	Strategic vacuum --- sound moves, no unifying plan
L5	Strong club player	1800--1999	Reactive play --- doesn't prevent the opponent's ideas
L6	Expert / CM	2000--2299	Conversion failure --- wins strategically, can't finish
L7	Master / Titled	2300+	Highly individual --- needs personalised diagnosis

2.3 The 5 plateaus

A plateau is a named stall pattern, auto-diagnosed from the *shape* of your skill vector — never self-reported. Zones deliberately overlap; ratings above 2200 have no formulaic plateau by design (that band is individual-coaching territory).

Plateau	Rating zone	Target skills
The blunder wall	0--1399	tactical pattern recognition, tactical consistency
The strategy desert	1400--1600	strategic planning, pawn endings, endgame principles
The conversion ceiling	1500--2000	converting advantages, rook endings, endgame precision
The prophylaxis gap	1700--1900	prophylaxis, defence & counterplay
The precision boundary	2000--2200	calculation precision, endgame precision, time mgmt, resilience

■ When a plateau is diagnosed

A plateau is only named when **(a)** your rating falls in its zone, **(b)** at least one of its target skills has real evidence, and **(c)** those target skills together carry at least **5** evidence samples. Among all qualifying plateaus, the one with the *lowest average mastery* across its evidenced targets wins. A brand-new account is given no diagnosis at all.

3 The Player's Journey

This section is the “how to play” walkthrough: the screens, in the order you meet them. The left sidebar groups them so the navigation itself tells you where you are: **Play** (Play vs AI) → **Your games** (Games, Library) → **Diagnosis** (Progress, Player Model) → **Training** (Training Plan, Drills).

★ FIRST TIME IN? TAKE THE TOUR

On your first sign-in a short **guided tour** walks you through the whole loop, and a “**Try a demo game**” shortcut analyses a famous master game so you see a real review in seconds — before you’ve uploaded anything. Both live behind the floating **?** button (bottom-right), which also opens the AI help assistant. See §3.12.

3.1 Register and sign in

Registration asks for a display name, an email, and a password of at least 8 characters. New accounts are created as role *player* on the *free* tier. Sign-in is rate-limited (20 attempts per 15 minutes per address) and returns a session token stored in your browser; it slides forward for 14 days of activity.

3.2 Play vs AI

No game to upload yet? Play one. Pick your colour and a difficulty from **Beginner** to **Max**, then play on an interactive board — click a piece, then a highlighted square. Your opponent is **Stockfish**, never the language model: at lower levels it searches shallower and sometimes picks a weaker (but always legal) move; at Max it plays the engine’s best every time. A **Coach** panel offers up to three graded hints for the position you’re facing — the same engine-grounded, can’t-hallucinate hints as Drills, so you get real in-match help without the model ever choosing a move for you.

When the game ends, **Save & analyse** runs it through the same pipeline as an uploaded game: it becomes one of *your* games, so it feeds your player model and generates drills (the opponent is labelled “Stockfish (Level N)” so practice games are easy to spot). This is the way to bootstrap a profile with no games to import — and, unlike a loaded library classic, a game you played *is* yours, so it counts.

3.3 The Dashboard (“Games”)

Your landing screen. Three stat cards summarise everything analysed so far — **Games analysed**, **Average accuracy**, and **Games with a blunder** — above a list of recent games. Each row shows the opening (ECO), both players with your side highlighted, the colour-coded result, time control, length, a blunder tag if any, and your accuracy. Accuracy is coloured green at 85+, gold at 75+, red below.

3.4 Upload games

Choose the source (Chess.com export, Lichess export, or manual PGN), tell the app **your username as it appears in the PGN** so it can detect which colour you played, then paste PGN text or drop a .pgn file. On submit, each game streams its analysis progress live. Unparseable games are listed with a reason and never sink the batch.

■ Upload limits & quotas

At most **50 games** per upload. Re-uploads of a game you already have are detected by content hash and cost nothing. Analysis is metered by tier: anonymous **1/day**, free **5/day**, Pro and Academy **unlimited**. Duplicate and over-quota games are never charged.

3.5 Game Review

The move-by-move board. On the left, an evaluation bar, the board oriented to your colour, and navigation (buttons or keyboard arrows / Home / End). Every move you step to shows a **classification badge**, the centipawns lost, how long you thought, the engine's preferred move, and the top three engine lines. On the right: live analysis progress, a **Game accuracy** card (White vs Black, with the engine depth and a tally of each classification), the full move list, and an **Engine summary** coaching note. *This note is the first place you meet the AI:* a language model writes it in plain English — but only from verified facts (your result, accuracy, and worst move), and it is re-checked by the claim guard before you see it, so it is badged “engine-grounded”. If the model is unavailable or its draft fails the check, you get a deterministic, engine-only summary badged “deterministic” instead. Either way, no sentence can contradict the engine (§1.2).

3.6 Player Model

Headed “*Every score has a receipt. No black box.*” On the left: your current level, your diagnosed plateau, an eight-axis skill radar, and a mastery-over-time sparkline. On the right: the full skill list with mastery bars and trend arrows, and an **Evidence** panel. Click any skill with evidence and it loads up to 20 receipts — the real moves behind the number, each marked as supporting the skill (✓) or counting against it (✗).

3.7 Progress — your guide to success

Headed “*Where you are, what you've achieved, and the single best thing to do next.*” This is the screen to open whenever you're unsure what to work on. It shows three things:

- a **Coach's summary** — *AI-written*, in plain language, grounded strictly in your real numbers (games analysed, evidenced skills, level, plateau, drill retention). It never invents a statistic; if the model is down you still get a deterministic summary of the same facts.
- **Your next move** — a single, concrete next action chosen by a fixed rule, not the AI: upload your first game → analyse enough to unlock a plan → clear today's drills → study your lowest well-evidenced skill. A **Go** button takes you straight there.
- **Key stats** and a **What you've done** list — deterministic, fact-by-fact, so the encouragement is always earned.

3.8 Training Plan

Headed “*Prescribed from your diagnosis, not a generic curriculum.*” It opens with **The hypothesis** — a sentence tied to your diagnosed plateau — then up to three **focus blocks**. Each names a weak skill, its mastery and evidence count, a rationale, and **matched reading** drawn from a book corpus mapped to the taxonomy. A skill needs at least **3** evidence points before it is allowed into a plan.

3.9 Drills

Headed “*Every position here is one you actually reached and misplayed.*” Each drill is one of your own mistake positions presented as a four-way multiple choice — the engine’s best move plus three legal decoys. Answer correctly and the card is scheduled further out; answer wrongly and it returns tomorrow. A stat block tracks cards **due**, your day **streak**, and **retention**. The scheduler is SM-2 spaced repetition (§4).

Stuck? Ask the Coach. A **Coach** panel offers up to three *graded, Socratic* hints: first a nudge to look at forcing moves, then the *kind* of move (a capture, a check, a quiet improvement), and only last the move itself. Crucially, these hints are *engine-grounded, not AI guesswork* — each is derived directly from the engine’s own best move, so a hint can never be wrong. Taking a hint is honest about what it means: a hinted-correct counts as **partial credit**, so the card comes back sooner than one you solved unaided (§4). Hints and help chats are recorded so your effort is transparent.

3.10 Library

Two tabs. **Games** is a searchable catalogue of master games — the golden-start library ships with roughly **6,000** games from Morphy, Capablanca, Fischer, Tal, and Kasparov, and grows via `npm run library:import`. You can **search** by player, opening, or event, **filter** by ECO code, result, and source, **sort**, and page through the results. Loading a game runs full engine analysis on it for review (one quota unit); because a historical game has no honest “your side,” a loaded classic is *not* folded into your player model — studying Morphy teaches you about Morphy, not about you. **Opening explorer** is an interactive tree: at every position it shows what the master library played there against what *you* have played there in your own analysed games — a direct read on where your repertoire leaks.

3.11 Account & data

GDPR controls. **Export** downloads every row keyed to you — games, moves, analyses, skill scores, evidence, coaching links — as a single JSON bundle (password fields stripped). **Delete** requires typing DELETE, then transactionally erases your data, tombstones the account, and logs you out. The deletion itself is recorded in an audit log for compliance.

3.12 Help, anytime

The floating ? button (bottom-right, on every screen) opens three ways to get unstuck:

- **Take the guided tour** — replays the eight-step walkthrough of the whole loop; it auto-runs once on your first visit and is always skippable.
- **Try a demo game** — loads a famous master game and analyses it immediately, with a short note explaining what you’re watching, so you can see a full review before uploading anything of your own.
- **Ask the help assistant** — a chat window backed by AI. Ask “how do I...” or “what does this mean” in plain language and it explains, referring you to the right screen. It answers *only* from a curated knowledge base about how Master Chess works, plus your own account context; it will not invent features or chess advice, and it hands you off to the Account page (or your administrator) for anything out of scope (§1.2).

4 The Mechanics

This is the rules reference: exactly how a game becomes a diagnosis. The worked example in §5 then runs all of it on one real game.

4.1 The pipeline, end to end

1. **Ingest.** The PGN is split (comment-safe), loaded move by move, and stored with before/after positions, clock times, and detected player colour.
2. **Evaluate.** Every position is scored by Stockfish (results are cached, since openings recur), producing an evaluation and ranked candidate lines for each.
3. **Classify.** Each move is labelled **Best** **Good** **Inaccuracy** **Mistake** **Blunder** by how much it lost against the engine’s best, scaled by game phase.
4. **Infer evidence.** Fixed rules read the classified moves and emit “for” / “against” evidence for specific skills.
5. **Update the model.** Evidence is folded into each skill’s Bayesian Knowledge Tracing state; a snapshot records level and plateau.
6. **Harvest & prescribe.** Mistakes become drills; the training plan is rebuilt; finally the narrator writes a claim-guarded summary.

★ FAILURE ISOLATION

The engine analysis is committed and shown to you *before* the enrichment step (evidence, model, drills, narration) runs. Enrichment is wrapped so that if any part of it fails, your analysis still stands as “done” — and it is idempotent, so a restart mid-way never double-counts evidence.

4.2 Move classification

A move’s raw loss is how many centipawns worse it is than the engine’s best reply. Thresholds are **phase-scaled** — the endgame is judged more strictly than the opening, where many moves are near-equal:

Phase	good >	inaccuracy ≥	mistake ≥	blunder ≥
Opening	5	60	150	350
Middlegame	5	50	120	300
Endgame	5	40	90	200

Values are centipawns of *effective* loss. At or below “good”, a move is **Best**.

Phase detection is queen-aware

Phase is not just move number. Very low material (a lone minor per side or less, ≤ 500cp) forces **endgame** even early — an early queen trade into a king-and-pawn position is an endgame. Otherwise the first 20 plies are **opening**; after that, low non-pawn material *with no queens on the board* (≤ 1400cp) is **endgame**; everything else is **middlegame**. Queens keep a position out of the endgame even when other material is thin.

Win-probability dampening

Raw centipawns mislead in decided positions: going from +900 to +700 is not a blunder, because you were completely winning before and after. The classifier converts evaluations to win probability on the Lichess curve and, *only when both endpoints sit deep in a decided zone*, blends the raw loss toward the much smaller win-probability loss. A genuine thrown win — say +2000 collapsing to +200 — is *not* dampened, because the second endpoint is no longer decided. Missing a forced mate is always at least a mistake regardless of centipawns.

■ Accuracy score

Per move, the app computes win-probability points lost, maps each to a 0–100 accuracy on the Lichess curve ($103.1668 e^{-0.04354x} - 3.1669$), then blends the arithmetic and harmonic means of the whole game. The harmonic term is what makes a single game-losing blunder cost more than a plain average would.

4.3 Turning moves into evidence

Rules fire only on *your* moves and emit evidence with a direction and a weight in $[0, 1]$. Error severity is graded blunder = 1, mistake = 0.6, inaccuracy = 0.3. A representative selection:

Rule	Fires when	Skill	Dir / weight
missed-forced-mate	you had a forced mate and gave it up	calculation precision	against / 1.0
opening-error	opening-phase mistake or blunder	opening principles	against / sev.
middlegame-blunder	a blunder in the middlegame	tactical consistency	against / 1.0
...blunder-pattern	(same move, also)	tactical pattern recog.	against / 0.5
hanging-piece-left	error that leaves a piece capturable & undefended	piece activity	against / 0.4
endgame-error	endgame mistake or blunder	endgame precision	against / sev.
snap-blunder	error played in $\leq 3s$	time management	against / 0.4
conversion-failure	reached $\geq +400cp$ but didn't win	converting advantages	against / 0.8
sustained-initiative	grew an even game to $\geq +500$ cleanly	attack & initiative	for / 0.55
held-worse-position	were ≤ -300 yet didn't lose	defence & counterplay	for / 0.5
consistent-repertoire	same opening, played cleanly, ≥ 3 games	opening repertoire	for / 0.4
sound-novelty	sound engine-approved deviation from book	opening prep.	for / 0.3
prophylaxis probe	a null-move probe reveals a real threat	prophylaxis	for/against

★ THE PROPHYLAXIS PROBE

To judge prophylaxis, the app performs a “null-move” probe: it lets your opponent move twice and asks the engine how much that would have cost you. If a real threat existed ($\geq 300cp$) and you let it land, that is evidence against your prophylaxis; if it existed and you neutralised it, that is evidence for. The weight is derived from the threat’s size, never from the language model. Probes are skipped when you were in check (the position would be illegal) and capped at six per game.

4.4 How each skill is scored

Every skill’s mastery comes from the same chain — *engine classifies your moves* → *a fixed rule turns specific patterns into for/against evidence* → *Bayesian Knowledge Tracing folds that evidence into a 0–100 number* (next subsection). What differs between skills is the **rule**: what, exactly, is measured. The Player Model shows this method in plain language for each skill, so a score is

never a mystery. Here is the full map.

Skill	What is measured
OPENING	
Opening principles	Errors in the first 20 plies count against; clean opening play counts lightly for.
Opening repertoire	Replaying the same ECO line in 3+ games with low opening cp-loss counts for. Variety is never penalised.
Opening prep & novelties	Leaving the master-game book (moves 4--10) with an engine-approved move counts for (sound novelties only).
MIDDLEGAME	
Tactical consistency	A middlegame blunder counts against --- the ``did you blunder-check?" measure.
Tactical pattern recognition	Middlegame mistakes/blunders where the engine clearly saw better count against.
Calculation precision	Missing a forced mate the engine had found counts heavily against.
Piece activity & coordination	Leaving your own piece attacked and undefended (on an error) counts against --- a real board check, not just cp-loss.
Attack & initiative	Growing an even position to clearly winning (+5) counts for.
Defence & counterplay	Being clearly worse (−3) yet drawing or winning counts for. A loss is not counted against.
Converting advantages	Reaching a won position (+400cp) but not winning counts strongly against.
Prophylaxis	A null-move engine probe gauges a position's standing threat; a real threat you let land counts against, one you neutralised counts for.
ENDGAME	
Endgame principles	Endgame mistakes/blunders count against; sound endgame play counts lightly for.
Endgame precision & conversion	Endgame cp-loss counts against --- precision once the board simplifies.
Pawn / rook / knight / bishop endings	The same endgame errors are <i>also</i> attributed to the specific piece-type ending they occurred in.
PSYCHOLOGY & MENTAL	

Skill	What is measured
Time management	A blunder/mistake played in under 3 seconds counts against.
Thought process & candidate moves	A blunder/mistake after 90+ seconds of thought counts against (had time, still missed it).
Psychological resilience	The spread of your cp-losses and whether errors spiked after your first blunder ("tilt") vs. stayed controlled.

■ "Not yet scored" is a real answer

Seven of the 27 skills have no honest rule yet, because they need judgment beyond move classification: *strategic planning, positional play, patience & long-term thinking, game transition, pattern-recognition speed, intuition & practical judgment, and psychological warfare*. Rather than show a made-up number, the app marks these **Not yet scored**. So a blank skill means "not measured yet" — never "you are bad at this."

4.5 Bayesian Knowledge Tracing: how a score moves

Each skill holds a probability p_{Know} that you have mastered it; mastery shown on screen is simply $\text{round}(p_{\text{Know}} \times 100)$. A new skill starts at $p_{\text{Know}} = 0.3$ (mastery 30). Each piece of evidence updates it with a weighted Bayesian step, using fixed parameters (slip 0.1, guess 0.2, learning transition 0.15).

■ The update, in words

A "for" observation raises p_{Know} toward the Bayesian posterior and adds a small learning bump; an "against" observation lowers it. **Weight scales the whole step**: a weight-1 blunder moves the number hard, a weight-0.3 inaccuracy nudges it, a weight-0 observation does nothing at all. Evidence is folded in sequence, so order within a skill matters.

The concrete effect of a *single* full-weight observation from a fresh skill (mastery 30):

Observation	New p_{Know}	New mastery
one "for", weight 1.0	0.710	71
one "against", weight 1.0	0.051	5
one "against", weight 0.3	0.225	23
weight 0 (any)	0.300	30 (no change)

Confidence in the whole model grows with the number of samples, capped: $\min(95, \text{samples} \times 3)$ percent.

4.6 Own-mistake drills and SM-2 scheduling

Every mistake or blunder of yours that lost at least 100cp — and for which the engine has a genuinely better move — becomes a drill. The correct answer is the engine’s own best move; positions are de-duplicated so you never drill the same one twice.

Scheduling is SM-2-lite. A card carries an *ease* factor (start 2.0, floored 1.3, capped 2.6) and an interval that grows multiplicatively so a correct answer can never shorten it.

■ The schedule

Correct: ease rises (more for better-known skills), and the next interval is $\text{interval} \times \text{ease}$, capped at **21 days**. **Incorrect:** ease drops by 0.2, the card returns in 5 minutes, and a lapse is recorded. After **4 lapses** a card is suspended as a “leech” and stops being scheduled. A card only counts toward your streak and retention when it was genuinely due — you cannot grind the same card for credit. **Hinted-correct** (you used the Coach) is partial credit: no ease reward, no streak, and only a gentle interval bump, so the card returns sooner than an unaided solve.

An answer is also accepted if it is any engine move within about 25cp of the best — two equally winning moves are both marked correct.

Correct answer #	Ease	Next interval	Meaning
1	2.08	2.08 days	graduates from ``new"
2	2.16	≈4.5 days	
3	2.24	≈10 days	
4	2.32	21 days (capped)	considered learned

Progression for a card on a mastery-30 skill, answered correctly each time.

5 A Complete Game of Play

We now run the whole system on one real game: Paul Morphy’s “Opera Game” (Paris, 1858), analysed as though *you uploaded it and played Black* — the allied Duke and Count. Every number, verdict, and skill change in the boxes below is the **actual output** of the Master Chess pipeline run on this game’s moves, not an illustration.

★ **THIS EXAMPLE WAS GENERATED, NOT HAND-WRITTEN**

The figures here come from running the app’s real classifier, evidence rules, and Bayesian update on the stored analysis of this game. One consequence worth stating up front: the model’s job is to read the moves honestly, and honest reading sometimes surprises — as it does here.

STEP 1 — UPLOAD & INGEST

You upload the game as Black. The ingester reads the header — White: Paul Morphy, Black: Duke Karl / Count Isouard, Result: 1-0, ECO: C41, Opening: Philidor Defense — matches your name to the Black seat, and replays all 33 plies, storing the position before and after each. The moves:

1.e4 e5 2.Nf3 d6 3.d4 Bg4 4.dxe5 Bxf3 5.Qxf3 dxe5 6.Bc4 Nf6 7.Qb3 Qe7 8.Nc3 c6 9.Bg5 b5 10.Nxb5 cxb5 11.Bxb5+ Nbd7 12.O-O-O Rd8 13.Rxd7 Rxd7 14.Rd1 Qe6 15.Bxd7+ Nxd7 16.Qb8+ Nxb8 17.Rd8#

A historical game carries no rating header, so — as §4 requires — no *level* or *plateau* is diagnosed from it. The skill evidence below is generated all the same.

STEP 2 — EVALUATE & CLASSIFY (ALL OF BLACK'S MOVES)

Stockfish scores every position; the classifier judges each Black move against the best available, phase-scaled. This is the complete, verbatim classification:

Move	Played	Phase	cp lost	Verdict
1...e5		opening	3	Best
2...d6		opening	9	Good
3...Bg4		opening	65	Inaccuracy
4...Bxf3		opening	54	Good
5...dxe5		opening	0	Best
6...Nf6		opening	67	Inaccuracy
7...Qe7		opening	18	Good
8...c6		opening	23	Good
9...b5		opening	84	Inaccuracy
10...cxb5		opening	139	Inaccuracy
11...Nbd7		middlegame	33	Good
12...Rd8		middlegame	58	Inaccuracy
13...Rxd7		middlegame	1	Best
14...Qe6		middlegame	132	Mistake
15...Nxd7		middlegame	mate	Blunder
16...Nxb8		middlegame	0	Best

Two things the real data teaches. First, the game is classified **opening** then **middlegame** throughout — the queen-aware phase rule keeps it out of the endgame despite the piece sacrifices. Second, the famous “losing” move **9...b5** is **only an inaccuracy** (84cp). The move the engine actually flags as the **blunder** is **15...Nxd7**, which walked into 16.Qb8+ Nxb8 17.Rd8# — a forced mate (shown as “mate” rather than a centipawn figure).

STEP 3 — INFERRED EVIDENCE (EXACTLY WHAT THE RULES EMIT)

The rule engine reads those classifications. A crucial detail visible here: **inaccuracies emit no per-move skill evidence** — only clearly good moves (**Best** / **Good**) and clear errors (**Mistake** / **Blunder**) do. So Bg4, Nf6, b5, cxb5, and Rd8 produce nothing. What actually fires:

Skill	From (real moves)	Dir / wt
Opening principles	e5, d6, Bxf3, dxe5, Qe7, c6 (six sound opening moves)	for / 0.15
Tactical consistency	Nbd7, Rxd7, Nxb8 (sound middlegame)	for / 0.12
Tactical consistency	15...Nxd7 (middlegame blunder)	against / 1.0
Tactical pattern recog.	14...Qe6 (middlegame mistake)	against / 0.6
Tactical pattern recog.	15...Nxd7 (blunder pattern)	against / 0.5

Note what does *not* fire: no “conversion failure” (Black never reached +400), no “held-worse-position” credit (Black lost), and no resilience verdict (the run did not meet the sample conditions). The model records only what the game shows.

STEP 4 — THE PLAYER MODEL UPDATES (REAL BAYESIAN OUTPUT)

Each skill folds its evidence in sequence from a starting mastery of 30. The actual results:

Skill	Net evidence	Mastery	Trend
Opening principles	6 for	30 → 66	↑
Tactical consistency	3 for, 1 against (wt 1)	30 → 12	↓
Tactical pattern recognition	2 against	30 → 9	↓

The surprise is honest and instructive: **opening principles rises to 66** even though Black *lost the game*. By cp-loss, Black’s opening was sound — the defeat came from two middlegame errors, not the opening. A human eye remembers “the game where I got mated”; the model measures each phase on its own evidence. That is precisely the black-box the receipts exist to open.

STEP 5 — DRILLS HARVESTED

Two moves qualify as own-mistake drills — classified **Mistake** or **Blunder** and losing at least 100cp:

Position before	Skill	Correct answer
14...Qe6 (132cp)	tactical pattern recognition	the engine's best move
15...Nxd7 (mate)	tactical consistency	the engine's best move

Each becomes a four-choice card, due immediately, entering SM-2 scheduling. Note that 9...b5 and 10...cxb5 are *not* harvested — b5 lost under 100cp, and both were inaccuracies

rather than outright errors.

STEP 6 — TRAINING PLAN PRESCRIBED

A skill needs at least 3 evidence samples to enter a plan. Here that admits *opening principles* (6 samples) and *tactical consistency* (4); with only 2 samples, *tactical pattern recognition* is honestly held back despite being the lowest score. Because the historical game has no rating, no plateau is diagnosed, so the plan uses its no-plateau hypothesis:

The hypothesis. *“Your weakest evidenced skills right now are tactical consistency, opening principles. No plateau pattern is diagnosable yet (needs a PGN-header rating), but these are the lowest-mastery skills we have real evidence for.”*

Focus block 1 — Tactical consistency (mastery 12, 4 samples). *Matched reading* from the corpus: *Art of Combination* (Blok), *Encyclopedia of Chess Combinations* (Chess Informant), *1000 Checkmate Combinations* (Henkin).

Focus block 2 — Opening principles (mastery 66, 6 samples), carrying its own matched titles. It ranks second only because it is the next evidenced skill — the plan is transparent about that rather than inventing urgency.

STEP 7 — THE ENGINE-GROUNDED NOTE

Finally the narrator is handed *only* verified facts — result, accuracy, blunder count, and the worst move (15...Nxd7, a mate-allowing blunder) — and asked for two to four sentences, then re-checked by the claim guard. A passing note reads like:

“Your opening was solid — it was the middlegame that cost you. 14...Qe6 let the pressure build, and 15...Nxd7 walked into a forced mate with 16.Qb8+ and 17.Rd8#. The habit to drill is the blunder-check before a recapture when your king is still in the centre.”

Every move and claim there maps to the real record. Had the draft said “your 9...b5 was the losing blunder” — contradicting the classifier, which graded b5 an inaccuracy — the guard would reject it and you would see the deterministic summary instead. *(That is the exact error this manual’s first draft made before it was checked against the engine.)*

★ WHAT YOU WOULD DO NEXT

Open *Drills* the next day: the 15...Nxd7 and 14...Qe6 positions are waiting. Choose the engine’s move and the card schedules out; miss it and it returns tomorrow until the gap closes. That is the loop — diagnose, explain, prescribe — closing on your own moves.

6 Reference Card

6.1 Screens at a glance

Play vs AI	play Stockfish at 5 levels; save & analyse to feed your model
Games	dashboard: stats + recent games
Upload	paste/drop PGN, live analysis
Game Review	per-move engine analysis, accuracy, AI summary
Progress	AI coach's summary + your single next move
Player Model	level, plateau, 27 skills, evidence receipts
Training Plan	hypothesis + 3 focus blocks + books
Drills	SM-2 review of your own mistakes
Library	searchable master games + opening explorer
Account	GDPR export / delete
? (help)	guided tour, demo game, AI help chat

6.2 Numbers worth remembering

Skills / categories / levels / plateaus	27 / 4 / 7 / 5
Fresh skill mastery	30
Blunder threshold (mid / open / end)	300 / 350 / 200 cp
Drill harvest threshold	100 cp lost
Drill accept tolerance	~25 cp of best
SM-2 ease (start / floor / cap)	2.0 / 1.3 / 2.6
Max review interval	21 days
Leech suspension	4 lapses
Plateau minimum samples	5
Plan minimum samples per skill	3
Free-tier analyses / day	5
Max games per upload	50

6.3 Classification legend

Best

Good

Inaccuracy

Mistake

Blunder